

Dyadic Pascal Skew Towers: An Exact Clock

Route-A structural certificate C166 — internal baseline

Abstract

For $q = 2^r$ and $d \geq 2$ we study the finite affine tower $T(x_1, \dots, x_d) = (x_1 + x_2, \dots, x_{d-1} + x_d, x_d + 1)$. Writing $a = \lfloor \log_2 d \rfloor$, finite evidence suggests the exact common clock $M = 2^{r+a}$. The central task is to prove this at every parameter from the Pascal iterate coefficients, rather than infer it from the released finite ledger. This baseline records the model pivot and theorem target; the valuation necessity proof and operator reversal are reserved for internal review.

Keywords: affine dynamics; Pascal coefficients; dyadic clock; finite orbit; skew tower; exact period.

1 Frozen model and target

Associate x with

$$p_x(t) = 1 + x_d t + x_{d-1} t^2 + \dots + x_1 t^d \quad \text{in } (\mathbb{Z}/q\mathbb{Z})[t]/(t^{d+1}).$$

Then T is multiplication by $1 + t$, whence

$$p_{T^n x}(t) = (1 + t)^n p_x(t), \quad T^n x = x \iff q \mid \binom{n}{k} \quad (1 \leq k \leq d).$$

The intended all-parameter result is

$$\text{Fix}(T^n) = \begin{cases} (\mathbb{Z}/q\mathbb{Z})^d, & M \mid n, \\ \emptyset, & M \nmid n, \end{cases} \quad M = 2^{r + \lfloor \log_2 d \rfloor}.$$

2 Model pivot and boundary

The standalone two-dimensional shear was not retained: its odd-modulus branch is product-rotation-like and its even branch is the $d = 2$ member above. The finite ledger is only a regression sentinel. No complexity theorem, target determinant match, arithmetic local datum, or Hilbert–Pólya operator is claimed. Scope: NO_BAD_EULER_OR_ROOT_NUMBER; Route B is unauthorized.

Declarations

No external dataset, human or animal subject, external funding, or known competing interest is involved. An AI coding assistant supported planning and drafting; it was not treated as an external reviewer.